Road Guard

Developed by: Rihan Mohamed K

Project type: Personal project

1. Introduction

Road Guard is an AI-powered computer vision system designed to **detect potholes** and road damage in real time as well as from images. The detection results are sent to a **Flutter-based** mobile application, where users can view the identified potholes along with relevant details. The app uses a Firebase backend to store, manage, and synchronize detection data in real time, ensuring efficient reporting and accessibility. The system aims to improve road safety by automatically identifying damaged roads and reducing accidents caused by poor infrastructure.

2. Problem Statement

Poor road conditions, especially potholes, lead to road accidents, vehicle damage, traffic disruptions, and increased maintenance costs. Despite these serious issues, road inspection processes are still mostly manual and slow. There is currently no widely implemented automated system for real-time pothole detection, making it difficult to identify and address road damage efficiently.

3. Proposed Solution

Road Guard uses a trained Convolutional Neural Network (CNN) model to analyze road images and classify them as either Clean Road or Pothole. The system displays the prediction results along with a confidence score to indicate the accuracy of detection. These results are then sent to a Flutter-based mobile application, where users can view the detected road condition in an organized interface. The application uses a Firebase backend to store detection data, manage user reports, and synchronize information in real time for efficient monitoring and future analysis.

4. System Architecture

A. Workflow:

i. Dataset Collection

a. Road images containing potholes were collected and organized.

ii. Model Training

a. The dataset was used to train a YOLO-based object detection model.
b. The model learned to detect potholes using bounding boxes.

iii. Real-Time Detection Script

a. A Python script was developed to load the trained YOLO model.
b. The system processes live camera feed or image input.

iv. Detection Output

a. The model identifies potholes in real-time.
b. The detection results are sent to the Flutter-based mobile application.
c. Data is stored and synchronized using a Firebase backend for real-time monitoring and record management.

5. Technologies Used

a. Python
b. Ultralytics YOLOv8
c. YOLOv8l Pretrained Model
d. Py-Torch
e. OpenCV
f. Custom Dataset with Bounding Box Annotations
g. GPU Acceleration (CUDA) for model training

6. Dataset Details

a. This project utilizes an online dataset collected from Kaggle.
b. Number of images of normal road – 412
c. Number of images of pothole – 780
d. Total No. Of images – 1192

7. Model Details

a. Trained on – RTX 4050 6gb (Laptop)
b. Epochs – 300
c. Batch size – 4
d. Optimizer – AdamW
e. Data augmentation techniques were used lip flip horizontally and vertically

8. Model Performance

The trained YOLO model achieved a precision of **65.3%**, indicating that most predicted potholes were correct. The recall was **66.7%**, showing that the model successfully detected a majority of potholes present in the dataset. The model obtained a **mean Average Precision (mAP@0.5) of 66.3%**, representing moderate detection accuracy at an Intersection-over-Union threshold of 0.5. A stricter metric, **mAP@0.5:0.95**, resulted in **38.3%**, reflecting the challenge of precise bounding box localization across multiple IoU thresholds.

9. ImgBB and Firebase implementation

When the YOLO model detects a pothole in a road image, the detected image is first uploaded to **ImgBB**, an image hosting service. ImgBB stores the image and generates a publicly accessible URL for the uploaded image. Instead of storing the full image in the database, only the **image URL provided by ImgBB** is stored in **Firebase**. This approach reduces storage usage in the database for free users.

10. Flutter Application

The Flutter mobile application connects to **Firebase** to retrieve the stored image URLs of detected potholes. When the application loads, it fetches these URLs from the database.

Using the retrieved URL, the Flutter application requests the corresponding image from ImgBB and displays it within the user interface. This allows users to view detected potholes directly inside the mobile application.

11. Project Presentations

The **Road Guard** system was successfully implemented and demonstrated. The complete system was deployed on a **Raspberry Pi 5** along with a webcam to enable real-time pothole detection using the trained YOLO model.

The project was presented at the **International Digital Fest**, conducted by **Cyber Square** and held at the **University of Dubai**. During the event, the system demonstrated the complete workflow of detecting potholes, uploading the detected images to ImgBB, storing the image links in Firebase, and displaying the results in the Flutter mobile application.

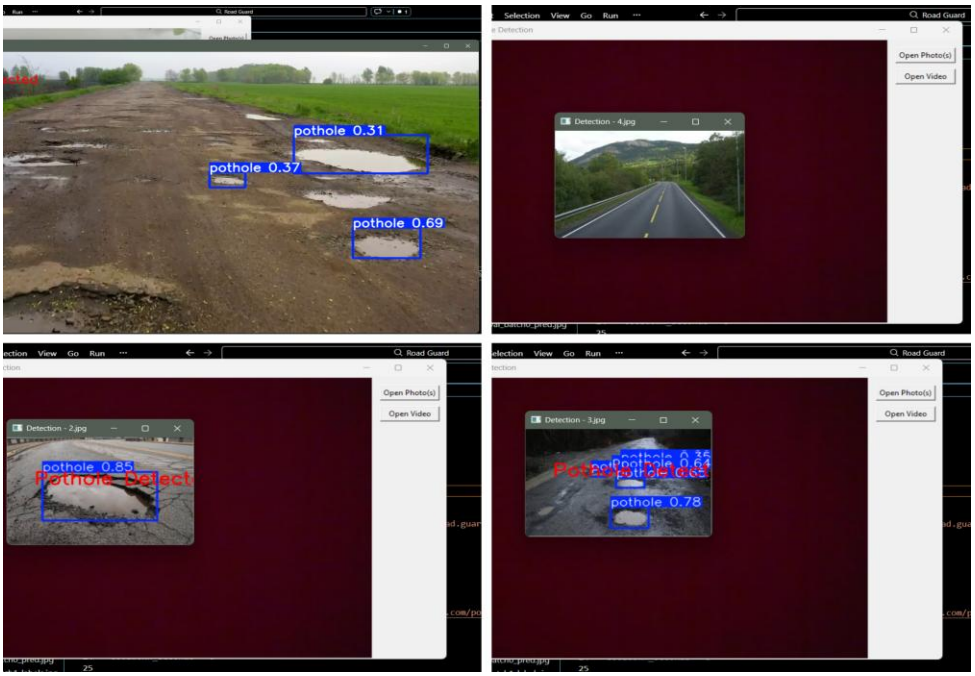
Among multiple participating projects, **Road Guard secured 3rd position** in the competition.

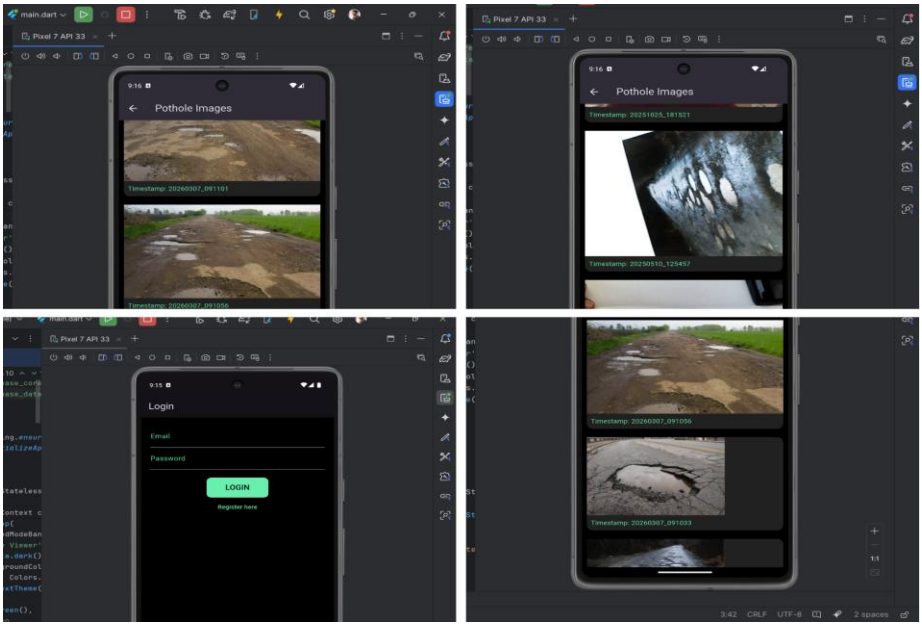
12. Future Work

In the future, the Road Guard system can be expanded by deploying it in local communities for real-time road monitoring. The system can be installed on vehicles such as **Auto rickshaws**, which regularly travel through different areas of a city or town. As these vehicles move through the roads, the system can continuously detect potholes and upload their exact locations and images to the database.

With support from government authorities, a centralized monitoring system can be created where detected potholes are automatically reported to the relevant departments. This would allow quicker identification and repair of damaged roads, helping improve road safety and infrastructure maintenance within the community.

13. Images




Pothole Images
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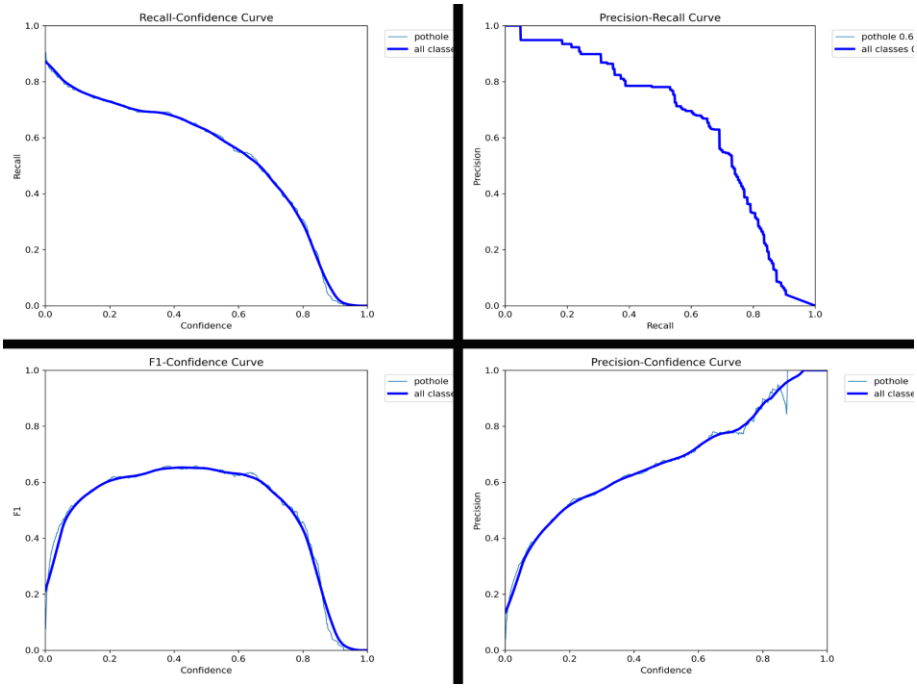
Register here


Recall-Confidence Curve
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Precision-Confidence Curve
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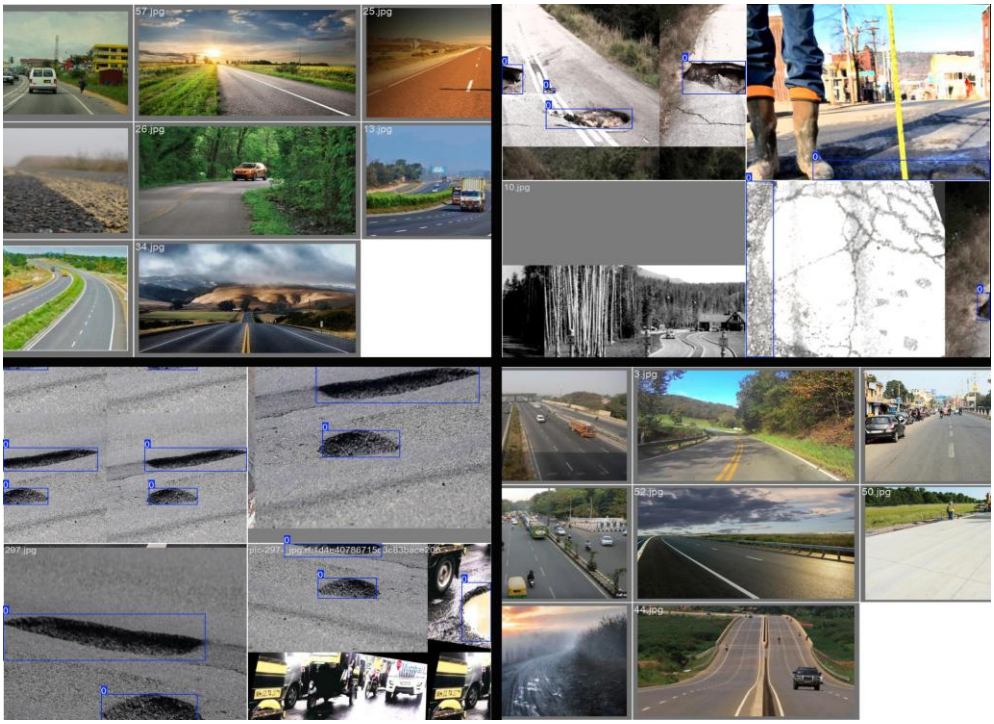
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